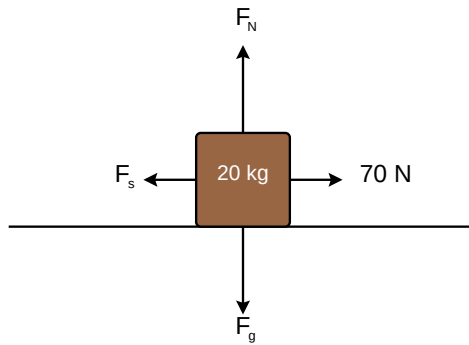


forces of friction - practice problems

name _____

1. A cafeteria tray slides across a counter. The coefficient of kinetic friction is 0.32, and the normal force on the tray is 75 N. What is the kinetic friction force?

2. A 20 kg crate sits on a level floor. The coefficient of static friction is 0.45. If you push horizontally with 70 N, does the crate move? What is the static friction force?



3. The same 20 kg crate has a coefficient of static friction of 0.45 and a coefficient of kinetic friction of 0.30. If you push horizontally with 100 N, will it move? If it moves, what is the kinetic friction force, and what is the crate's acceleration?

4. A 10 kg box slides on a level floor, and the friction force on it is 34.3 N. What is the coefficient of kinetic friction?

5. A student uses a spring scale to pull a wooden block across a level lab table at a constant speed. The scale reads 3.92 N. The coefficient of kinetic friction is 0.40. What is the mass of the block?
6. An 8.0 kg crate is pulled across a level floor with a 40 N horizontal force. The coefficient of kinetic friction is 0.20. What is the crate's acceleration?
7. A kid on a sled has a total mass of 30 kg. They reach the flat bottom of a hill moving at 18 km/h. The coefficient of kinetic friction between the sled and the snow is 0.10. How long does it take the sled to stop?
8. A 10 kg box slides across a warehouse floor at 4.0 m/s. What is the kinetic friction force on the box?
9. A 1400 kg car is driving at 20 m/s on a dry concrete road when the driver slams on the brakes and the tires lock up and skid. Use a coefficient of kinetic friction of 0.70 for rubber on concrete (from the table on this page). How far does the car skid?

10. A 6.0 kg block sits on a level table. The coefficient of static friction is 0.40. The block is attached by a string over a pulley to a 2.0 kg hanging mass. Will the system stay at rest? If so, what is the static friction force on the table block?

11. A 20 kg sled is pulled across level snow by a rope with 60 N of tension at 30° above the horizontal. The coefficient of kinetic friction is 0.10. What is the sled's acceleration?

12. A brick slides across a floor lying on its largest face. Then it is turned on its narrow side, so much less of it touches the floor. How does the kinetic friction force change?

13. A 10 kg box rests on a 15° ramp and does not slide. What are the normal force and the static friction force holding it in place?

14. A 10 kg box is placed on a 25° ramp. The coefficient of static friction is 0.35. Will the box stay at rest or slide?

15. To measure friction, a student slowly tilts a wooden board with a small block on it. The block starts to slide when the board reaches 31° . What is the coefficient of static friction between the block and the board?

16. A 12 kg box slides down a 20° ramp. The coefficient of kinetic friction is 0.20. What is the box's acceleration down the ramp?

17. A block slides down an 18° ramp with an acceleration of 1.2 m/s^2 . What is the coefficient of kinetic friction?

18. A 25 kg crate starts at rest on a level storage room floor. The coefficient of static friction is 0.50 and the coefficient of kinetic friction is 0.35. A person pulls horizontally with 150 N for 3.0 s. How far does the crate move?

19. A force sensor slowly pulls harder and harder on a 2.0 kg block resting on a level table. The table shows the friction force at each pull. Find the coefficients of static and kinetic friction.

applied force (N)	friction force (N)	block
0	0	at rest
4.0	4.0	at rest
8.0	8.0	at rest
11.8	11.8	just starts to slide
12.0	7.8	sliding